Math 526, S05 **Name:**

Quiz 7 **SSN: xxx-xx-**

<u>Instructions:</u> **There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (20 points) Consider the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & -4 & 2 \\ 2 & -1 & 4 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} -1 \\ -1 \\ 5 \end{bmatrix}$$

a) Write down the corresponding $\mathbf{S}$, $\mathbf{T}$ and the iterative formula for the Jacobi method.
Solution: The Jacobi method:

$$\mathbf{S}\mathbf{x}^{new} = \mathbf{b} + \mathbf{T}\mathbf{x}^{old}$$

$$\mathbf{S} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \qquad \mathbf{T} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & -2 \\ -2 & 1 & 0 \end{bmatrix}$$

i.e.,

$$\begin{aligned} x_1^{new} &= (-1 - x_2^{old})/2 \\ x_2^{new} &= (-1 - x_1^{old} - 2x_3^{old})/(-4) \\ x_3^{new} &= (5 - 2x_1^{old} + x_2^{old})/4 \end{aligned}$$

b) Starting with the initial guess $\mathbf{x}_0 = \mathbf{0}$, compute $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ using the Jacobi method. Also compute $||\mathbf{x} - \mathbf{x}_3||_\infty$ where $\mathbf{x} = [-1, 1, 2]^t$ is the exact solution.
Solution: Using $\mathbf{S}\mathbf{x}_{k+1} = \mathbf{b} + \mathbf{T}\mathbf{x}_k$ from a), we get

$$\mathbf{x}_1 = \begin{bmatrix} (-1-0)/2 \\ (-1-0-0)/(-4) \\ (5-0+0)/4 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/4 \\ 5/4 \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} (-1-1/4)/2 \\ (-1+1/2-5/2)/(-4) \\ (5+1+1/4)/4 \end{bmatrix} = \begin{bmatrix} -5/8 \\ 3/4 \\ 25/16 \end{bmatrix}$$

$$\mathbf{x}_3 = \begin{bmatrix} (-1-3/4)/2 \\ (-1+5/8-25/8)/(-4) \\ (5+5/4+3/4)/4 \end{bmatrix} = \begin{bmatrix} -7/8 \\ 7/8 \\ 7/4 \end{bmatrix}$$

Then it is easy to find that

$$||\mathbf{x} - \mathbf{x}_3||_\infty = 1/4$$